

[illegible][illegible]

Status Indicators

The diagram illustrates a circuit for status indicators using an RGB LED (D2). The LED is connected to a 3.3V_ESP supply. The circuit includes three resistors (R11, R12, R13) and three text labels (LED_RED, LED_GREEN, LED_BLUE) to indicate the status of the LEDs. The resistors are labeled 1k, and the text labels are labeled Text.

Environmental Sensing & Display

The diagram illustrates the hardware setup for environmental sensing and display. It features three main components: an ESP8266 microcontroller, an SSD1306 display module, and a DHT22 digital temperature and humidity sensor.

ESP8266 Connections:

- Power:** The `3.3V_ESP` pin is connected to the `VCC` pin (pin 01) of the J3 header. The `GND` pin is connected to the `GND` pin (pin 02) of the J3 header.
- I2C:** The `I2C_SCL` pin is connected to the `SCL` pin (pin 03) of the J3 header. The `I2C_SDA` pin is connected to the `SDA` pin (pin 04) of the J3 header.
- Other:** The `CSB` pin is connected to the `CSB` pin (pin 05) of the J3 header. The `SDD` pin is connected to the `SDD` pin (pin 06) of the J3 header.

SSD1306 Display Connections:

- Power:** The `GND` pin is connected to the `VDD` pin (pin 1) of the J4 header. The `3.3V_ESP` pin is connected to the `VDD` pin (pin 2) of the J4 header.
- I2C:** The `I2C_SCL` pin is connected to the `SCK` pin (pin 3) of the J4 header. The `I2C_SDA` pin is connected to the `SDA` pin (pin 4) of the J4 header.

DHT22 Sensor Connections:

- Power:** The `VCC` pin is connected to the `3.3V_ESP` pin.
- Signal:** The `SDA` pin is connected to the `DHT22` pin (pin 2) of the J6 header.
- Ground:** The `GND` pin is connected to the `GND` pin (pin 4) of the J6 header.

Capacitor: A `1uF` capacitor (C12) is connected between the `3.3V_ESP` pin and the `GND` pin to provide decoupling for the power supply.

User Input Interface

UART Programming

Diagram illustrating the UART Programming setup for an ESP module:

- ESP_TX (Pin 1)
- ESP_RX (Pin 2)
- ESP_RST (Pin 3)
- ESP_BOOT (Pin 4)
- SW4 (Switch)
- Sw (Switch Label)
- GND (Ground)



